Transcript: What's the problem?

Wilderness for Sale

Slide 2: Video Topics

- ➢ (Over)Population
- ➢ Carrying Capacity
- ➢ (Over)Consumption

This is the list of topics that will be covered in this video lecture. Population, or overpopulation, carrying capacity, and consumption or overconsumption.

Slide 3: (Over)Population

- ➢ World population growth
  - growth rate decreasing
    - the sum of the difference between births less deaths and the difference between immigrants less emigrants, for a country over one year (at the global level there is no migration)
  - fertility rates decreasing
    - average births per woman, > 2.1 is needed for growth
  - global population is still increasing, but more slowly
    - population growth is exponential

Near the turn of the nineteenth century, the world population reached a new milestone: one billion people. From there, it took more than a century to achieve two billion people and then far less time to triple that. Today, the world's population has surpassed eight billion people. And that's a whole lot of people.

So, what does the present look like and what does the future look like in terms of global population, growth, and decline?

The first thing I want you to know is that the population growth rate has been decreasing since it hit its peak in 1968. Today, the population growth rate is less than 1%. Keep in mind that this is the global rate of natural increase, births minus deaths since there is no migration at the global scale.

Also decreasing is the average number of births per woman. For a population to grow, this needs to be above 2.1, which is considered the replacement level. Below that, and at a global scale, the population will age and eventually decline. Many countries today have fertility rates below the replacement rate and extremely low rates of natural increase, but immigration keeps the populations of some of those countries growing.

Now let's talk about sheer numbers, because again, over 8 billion and growing means a lot of people. The one hundred years between 1920 and 2020 was a time of incredible growth and the global population nearly quadrupled.

Even with a slowing growth rate, population growth is exponential. The larger the base the greater the growth. Two people have two children, and those two children each have two children, and those four children each have two children, and on and on it goes.

Looking ahead to the end of this century, the global population is only expected to be about 25% larger, so somewhere around 10 billion in 2100, which is nowhere close to doubling.
The bottom line is that the global population is large, and increasing, but much more slowly than in the past.

Slide 4: Images "Comparison of United Nations population projections, Word" and "World population by region, including UN projections"

Population explosions are temporary, and we have seen this at the country level.

For many countries, that period of rapid growth has already ended and soon that will be the case for the world. The global fertility rate was cut in half between 1970 and 1920, which is a good indicator of where things are headed.

Up until 1700 – high childhood mortality counteracted high fertility and the population grew very slowly.

The Industrial Revolution happened, the Medical Revolution happened, health improved and mortality declined. At that point, things changed quickly, first in the early-industrialized countries and then spreading to the rest of the world. Between about 1950 or so and 2020, which is the steepest part of that graph on the right, the global population more than tripled, with a significant number of people–more than 3.5 billion–being added in the latter half of the twentieth century.

In the two centuries between 1800 and 2000, there was a 7-fold increase in population and this is when the greatest environmental change took place.

Technological advances coupled with explosive population growth amplified humanity's impact on the natural environment. In the 20th century we saw rivers burn, oil spill, smog kill, and black rain. The environment was sounding an alarm.

Providing space, food, and resources for an incredibly large population for decades to come, while also minimizing or mitigating damage, is one of the biggest challenges we face today.

We are still adding tens of millions of people to the population every year and we must find a way to provide for them without completely disabling the environment's ability to do so.

The United Nations projects that the global demographic transition will end, population growth will continue its decline and the curve representing the world population will level off. It is expected that the population growth rate will fall to 0.1% and from there, who knows? The population projection, the medium variant, is just a little more than 10 billion people in the 2080s before it levels off and then declines.

A lot of this is dependent on fertility rates, and fertility rates decrease when economic and human development increase.

You can see in the graph on the right that most of the greatest growth moving forward will happen in countries of Africa and Asia, however, Asia, with an incredibly large base population, has seen some pretty significant decreases in fertility rates.

Slide 5: (Over)Population

- ➢ Characteristics of growth today
  - population growth (and fertility rates) is highest in the least developed countries (UN Classification)
  - population growth rate is negative in some countries (notably Japan and Germany)
  - in some countries, immigration accounts for more than half of population growth

Keeping with that thought, the world fertility rate is falling, but it remains high in many developing countries. According to the United Nations, just eight countries will account for more than half of the world's population growth between now and 2050, five in Africa and three in Asia.

This will put a huge strain on resources in those countries, not just necessities like food and water, but housing, schools, and utilities as well. Plus, government programs and policies that focus on things like reducing poverty, improving healthcare, and sustainability, face even greater pressure.

Life expectancy has increased, and infant mortality has decreased, but it is fertility and family size that are driving growth in most countries.

The base population is also a factor, India and Nigeria are two of the most populous countries on the planet. Nigeria's fertility rate remains exceptionally high, while India's has fallen below replacement level. Even if fertility rates were to fall, as they have in India, it takes decades before that is reflected in the population.

Other countries are experiencing negative population growth or decline. Japan is one country whose population has been on the decline for some time now. China is facing a similar situation, and many other countries have stable, declining, or soon-to-be declining numbers as well.

In Canada, immigrants account for almost three-quarters of population growth, and in the United States, even more than that.

Slide 6: Images of population growth, food security, fertility and poverty

Our World in Data is a great resource for charts, visualizations, maps, and online or embedded, they are also interactive. And you'll notice I use them quite a bit.

These maps are not current, but they do paint a picture. Population growth has slowed dramatically in many countries, but it is still high in a handful of countries, and many of those countries will account for much of the growth in the decades to come.

Many countries with growing populations are considered least developed by the UN classification system, and if you are unsure what that entails, I encourage you to look it up.

Some of the countries with the fastest-growing populations are also those with the highest poverty rates, and many of those living in poverty are found in rural areas. It is estimated that 80% of the world's poor live in rural areas and the poverty rate in rural areas is more than three times higher than it is in cities.

These people are those most dependent on the environment for their subsistence needs and livelihood. They are heavily reliant on ecosystem services, highly vulnerable to the negative impacts associated with climate change, and those least able to do something about it.

Now, that is not to say that more developed countries don't have poverty or food insecurity issues. And really, you could argue that in a high-income or rich nation, poverty and food insecurity should be close to nothing, but of course, that is not the case.

Slide 7: (Over)Population

- Thoughts on growth & the environment
    - Neo-Malthusianism
        - uncontrolled population growth will lead to ecological collapse and must be curbed
    - Cornucopianism
        - technology and human ingenuity will save us
    - Conflict Theory
        - humans are in competition for resources, the issue is not a lack of resources but the inequitable distribution of those resources

Now I am going to transition into a conversation about population size and resource use. When the population is viewed as a factor of production, people are labor and more labor equals more production. That is assuming